Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

MSCS candidate at USC with 2+ years shipping full-stack web products in **TypeScript**, **React**, **Node.js**, and **Python** on **PostgreSQL** and **MySQL** backends, focused on performance, reliability, and user impact.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Analysis of Algorithms, Operating Systems

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, OOP, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Shipped user-facing platform features on **FastAPI** and **PostgreSQL** for 500+ users with **CI/CD** releases, improving dashboard latency **1.2s to 350ms** through schema design, query tuning, and automated validation checks.
- Built integration workflows processing 5K+ monthly jobs with reconciliation and monitoring hooks, maintaining 99%+ reliability while documenting changes for cross-team handoffs and faster production triage.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Built full-stack AI grading product with **React** and **Flask** on **AWS/MySQL**, serving 150+ instructors and 10K+ daily requests with **GitHub Actions CI/CD** and stable peak-load performance.
- Owned onboarding-to-feedback flows from conception through production, collaborating in code reviews to reduce turnaround **60%** while independently debugging live issues and improving maintainability.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Improved API performance **30%** through pagination and query optimization for low-bandwidth users while maintaining 20+ **PyTest** suites and **Jenkins CI/CD** across Linux, Windows, and macOS in Agile sprints.
- Collaborated with Microsoft/GitHub on integration testing and nightly automation, boosting pipeline efficiency **68%** and reducing manual validation through documented release workflows.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed **MERN** CRM with **React**, **Node.js**, and **WebSockets** syncing 2M+ records on **Azure SQL** with relational schema design and query tuning for high-volume workflows.
- Automated **Jenkins CI/CD** and Docker deployments for eight microservices, cutting release time **50%** while improving reliability via **Sentry** error tracking and performance monitoring for production triage.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Built **Flask APIs** and booking workflows for 50K+ records with sub-200ms responses, collaborating across 12+ endpoints on integration requirements and documenting system behavior for cross-platform delivery.
- Reduced manual refund handling **70%** by diagnosing production defects through logging and monitoring while independently unblocking integration issues and supporting rapid iteration cycles.

Skills

Languages: TypeScript, JavaScript, Python, SQL, Java, HTML/CSS, Bash
Frontend & Web: React, Next.js, TailwindCSS, responsive UI, WebSockets, REST APIs
Backend & Data: Node.js, FastAPI, Flask, PostgreSQL, MySQL, MongoDB, Azure SQL, microservices
Tools: Git, Docker, AWS, Azure, Jenkins, GitHub Actions, CI/CD, Sentry, Agile
Practices: code reviews, PyTest, automated testing, performance optimization, collaboration, documentation

Projects

FinNexus Live | *Next.js, TypeScript, MongoDB, WebSockets*

- Built scalable fintech web platform on **Next.js** and **TypeScript** for 50K+ users and 1M+ transactions with real-time updates, secure access controls, and performance-tuned **MongoDB** data access.

PiXMod Live | *Next.js, MongoDB, Stripe, SaaS*

- Shipped subscription SaaS on **Next.js** and **MongoDB** with Stripe billing, responsive UI, and semantic search across 100K+ assets while optimizing onboarding and premium adoption flows.

CPU-scheduling Simulator Live | *C++, React, Systems Programming*

- Implemented performance-critical **C++** scheduling logic with a **React** interface, handling edge cases and optimizing resource allocation across 1K+ simulated process combinations for interactive exploration.